$$\omega = \frac{Ba}{\sqrt{mL}}, \tag{18}$$

. :

$$x_0 = \frac{FL}{B^2 a^2}. \tag{19}$$

F

$$x_0 = a/2, \tag{20}$$

,

$$F_{\min} = \frac{B^2 a^3}{2L} = 5.00\times10^{-5}\ \text{H}. \tag{21}$$

7. 6

:

$$t_0 = \frac{\pi}{\omega} = \pi\frac{\sqrt{mL}}{Ba} = 7.02\ \text{c}. \tag{22}$$

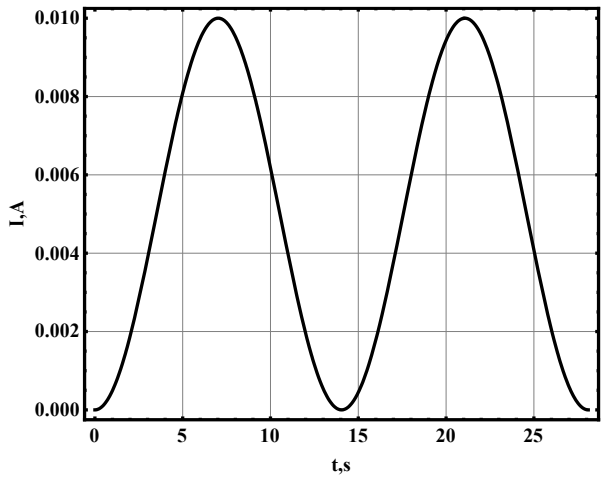
8. (17) $x(0)=0, x'(0)=0$:

$$x(t) = \frac{F_{\min}L}{B^2a^2}\left(1-\cos\omega t\right) = \frac{a}{2}\left(1-\cos\omega t\right). \tag{23}$$

(15) :

$$I(t) = \frac{Bax(t)}{L} = \frac{Ba^2}{2L}\left(1-\cos\omega t\right). \tag{24}$$

:



9. :

$$Ba\mathrm{v} - L\frac{dI}{dt} = IR, \tag{25}$$

:

$$m\frac{d\mathrm{v}}{dt} = -BIa. \tag{26}$$

(25) (26) :

$$Ba^2 - LI_0 = qR, \tag{27}$$

$$m\mathrm{v}_0 = Bqa, \tag{28}$$

q .

(27) (28) :

$$I_0 = \frac{B^2a^3 - m\mathrm{v}_0R}{aBL}. \tag{29}$$